

Winding-Free Exact Diagonalization of Quantum Spin Ice

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Quantum spin ice realizes an emergent three-dimensional $U(1)$ gauge theory whose low-energy dynamics is generated by six-spin ring exchange around pyrochlore hexagons. The sign of this exchange selects zero- or π -flux gauge backgrounds, yet the frustrated sign relevant to π -flux candidates is inaccessible to sign-problem-free quantum Monte Carlo. Exact diagonalization is sign unrestricted, but standard periodic clusters admit a noncontractible four-spin process at second order, parametrically above the physical third-order hexagon tunneling. We eliminate this finite-torus artifact by coupling microscopic spin transfers to a source that records their transport on the covering lattice and retaining only the zero-transport component. We show that this projection requires no reference frame and no spectrally isolated band: it is realized exactly by the Feshbach map onto the combinatorial ice manifold, and is realized as a genuine Hermitian operator $H_0 + C$ by a counterterm supported inside that manifold, which cannot dress itself and therefore enters the effective theory undressed. Such a counterterm is exact at a single energy, and we show that the energy is not a free parameter but is fixed by a Brillouin–Wigner condition whose solution places the exact point at the ground state—so that the winding-free ground multiplet, including the degeneracy that the transport-sector decomposition forces on it, is reproduced exactly rather than approximately. Solving that condition to 10^{-13} , rather than iterating it a few times, is the whole content of the method: the splitting of the protected multiplet, together with the spurious low-temperature anomaly and violated entropy sum rule it produces, is proportional to the residual over ten decades and is not intrinsic. The converged construction needs no fitted correction and satisfies the sum rule identically; its ground energy is the lowest exact masked root to $10^{-12}J_{zz}$, and it places the ring-exchange anomaly within 0.2% of the frame-free reference throughout $|J_{\pm}|/J_{zz} \leq 0.18$ while extending past the coupling at which that reference ceases to exist. Across $-0.30 \leq J_{\pm}/J_{zz} \leq 0.05$ the winding-free heat-capacity maximum follows $T_{\text{peak}} \propto |J_{\pm}|^{3.10}$, against $|J_{\pm}|^{1.99}$ for ordinary periodic exact diagonalization, separating the physical scale $12|J_{\pm}|^3/J_{zz}^2$ from the finite-torus scale $4J_{\pm}^2/J_{zz}$. The residual limitation is not the construction but the weight of the true state on the ice manifold, which we measure and which governs every failure mode of every variant we tested.

Introduction.—Quantum spin ice (QSI) is a $U(1)$ quantum spin liquid with fractionalized excitations and an emergent gauge field whose gapless photon mode provides a characteristic low-energy signature [1–3]. Recent experiments on $\text{Ce}_2(\text{Zr},\text{Sn},\text{Hf})_2\text{O}_7$ suggest that Ce-based pyrochlores may realize a π -flux QSI, in which an emergent gauge flux of π threads each elementary hexagonal plaquette [Fig. 1(i.a)] [4–6]. In $\text{Ce}_2\text{Zr}_2\text{O}_7$, no long-range order or spin freezing is observed down to the lowest accessible temperatures. Its specific heat features one broad hump, whereas recent heat-capacity measurements on $\text{Ce}_2\text{Hf}_2\text{O}_7$ resolve two anomalies near 65 mK and 25 mK [7]. Numerical linked-cluster fits on the high-temperature tail of the specific heat measurement place the candidate parameters within the π -flux QSI regime, which is a regime where the average flux threaded through a plaquette is π . But a true microscopic investigation of the origin of the two-peak structure is still lacking, particularly on the low-temperature peak.

This is because obtaining a microscopic description of π -flux QSI using unbiased numerical methods is challenging. Quantum Monte Carlo (QMC) simulations access the zero-flux QSI [8–10] but encounter a sign problem in the π -flux regime. Numerical linked-cluster expansions fail to converge at the ultralow temperatures at which the lower anomaly emerges. Finite-temperature

tensor-network calculations have reached a periodic 48-site three-dimensional Heisenberg cluster, but only well above the ring-exchange scale [11]. Density-matrix renormalization group and minimally entangled typical thermal states reach lower temperatures on pyrochlore tubes [12], but their quasi-one-dimensional geometry cannot support a gapless photon mode. A sign-unrestricted microscopic treatment that resolves the three-dimensional gauge dynamics at temperatures of order g_6 is therefore needed.

Exact diagonalization (ED) is precise on a finite cluster and imposes no sign restriction, but is limited to small pyrochlore tori, notably the cubic 16-site and face-centered-cubic 32-site clusters. Their dominant artifact is not size alone but the appearance of spurious four-site loops. To see this, we first introduce the physics of the gauge sector dynamics. The minimal microscopic setting for QSI is the nearest-neighbor pseudospin-1/2 XYZ model

$$H_0 = J_{zz} \sum_{\langle ij \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^- + S_i^- S_j^+) + J_{\pm\pm} \sum_{\langle ij \rangle} (S_i^+ S_j^+ + S_i^- S_j^-), \quad (1)$$

with $J_{zz} > 0$. For $|J_{\pm}|, |J_{\pm\pm}| \ll J_{zz}$ the low-energy states obey the spin-ice rule $Q_t \equiv \sum_{i \in t} S_i^z = 0$ on every

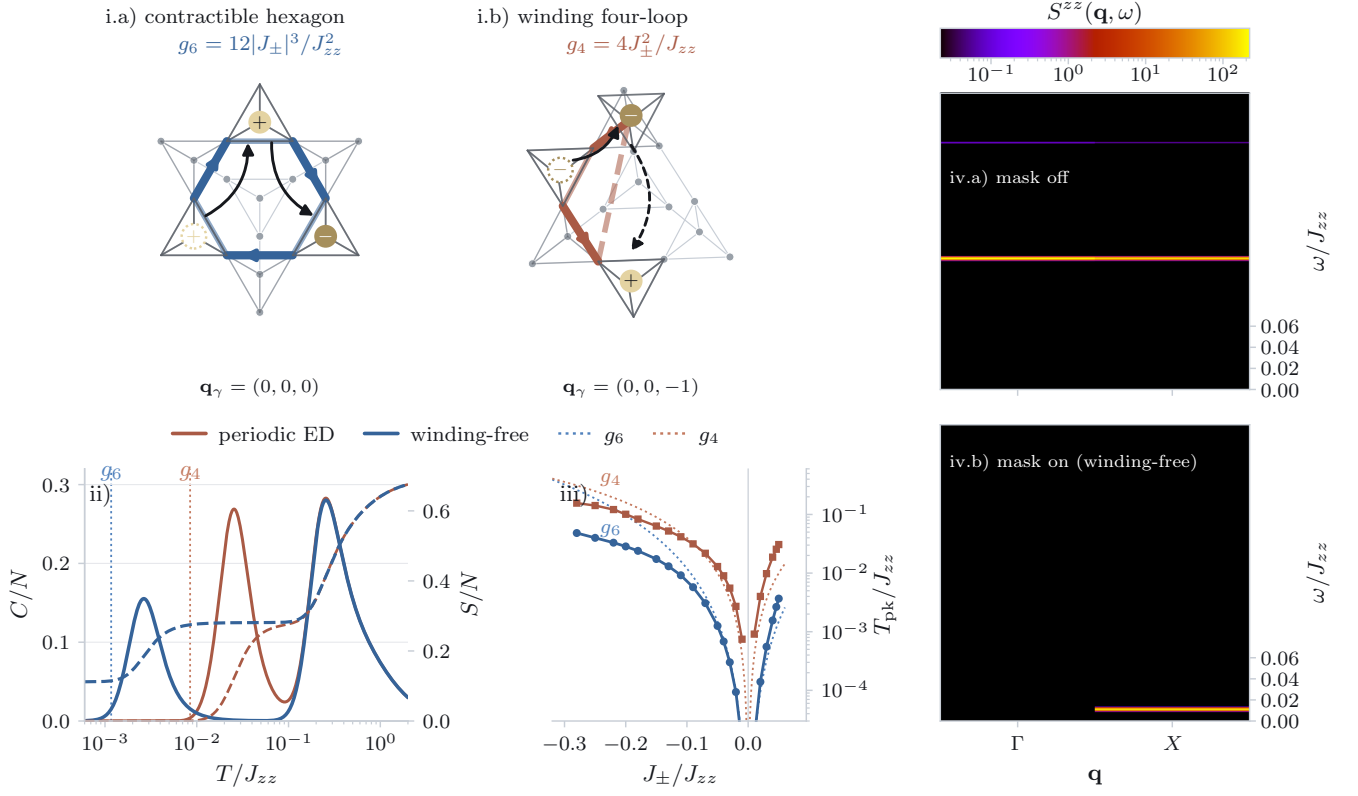


FIG. 1. Microscopic transport and winding removal on cubic-16. (i.a),(i.b) The two shortest ice-to-ice processes on the 16-site torus. Gray marks the cluster; colored circles and bonds are the loop, with its exchanges numbered in order and each drawn as an arrow from the lowered to the raised spin. An exchange leaves the tetrahedron it straddles neutral and charges the two outer ones, so it creates, moves, or annihilates a spinon–antispinon pair; charges sit at the centre of their tetrahedron, whose frame is drawn, light for $Q_t > 0$ and dark for $Q_t < 0$, faded where a charge has since left. The printed triple below each panel is the total transport $\mathbf{q}_{\gamma}$ of Eq. (5). (i.a) On the contractible hexagon the pair is created, walks, and annihilates inside one cell: $\mathbf{q}_{\gamma} = \mathbf{0}$, and the process enters at third order, at the g_6 of Eq. (2). (i.b) The winding four-loop closes only through the periodic identification (dashed bond): it transports one unit of polarization across the cell, $\mathbf{q}_{\gamma} = (0, 0, -1)$, enters already at second order, at the g_4 of Eq. (3), and is removed by the mask of Eq. (6), which retains the hexagon. (ii),(iii) One color code and one legend serve both panels: periodic ED (orange), the winding-free construction (blue), and the dotted perturbative scales g_6 and g_4 , labeled in each panel. (ii) Heat capacity (solid, left axis) and entropy (dashed, right axis) at $(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0)$, each from the complete 65536-level spectrum of the same cluster: the periodic low-temperature peak sits at the artifact scale g_4 , whereas the winding-free peak moves down to the ring-exchange scale g_6 . (iii) Position of the low-temperature heat-capacity maximum across the coupling range: the winding-free peak follows g_6 as $J_{\pm} \rightarrow 0$ on both flux sides, while the periodic peak tracks g_4 throughout. (iv.a),(iv.b) Zero-temperature longitudinal structure factor $S^{zz}(\mathbf{k}, \omega)$ of the downfolded ice-block dynamics at a common anchor, without and with the mask, on one logarithmic color scale, averaged over the three symmetry-related X points with broadening $\eta = 6 \times 10^{-4} J_{zz}$. Because S^z is diagonal in the Ising configuration basis, $P_{\text{ice}} S_{\mathbf{k}}^z P_{\text{ice}}$ is exactly a diagonal matrix and both panels are the same 90×90 eigenvalue problem fed identical sources: they differ only by whether Eq. (6) was applied, so the figure isolates the artifact rather than the downfolding. No reference frame enters either one.

tetrahedron t ; we write P_{ice} for the orthogonal projector onto that manifold, of rank 90 on cubic-16. Transverse exchange first connects distinct ice configurations at third order, through collective flips around elementary six-site hexagons, generating the ring-exchange scale [Fig. 1(i.a)]

$$g_6 = \frac{12|J_{\pm}|^3}{J_{zz}^2}. \quad (2)$$

In the convention of Eq. (1), $J_{\pm} > 0$ gives the zero-flux QSI, whereas $J_{\pm} < 0$ gives the π -flux QSI. The paramet-

rically small scale g_6 is therefore a natural candidate for the lower-temperature thermodynamic anomaly.

On cubic-16, a noncontractible four-spin path closes through the periodic boundary [Fig. 1(i.b)] and contributes already at second order, with

$$g_4 = \frac{4J_{\pm}^2}{J_{zz}}, \quad \frac{g_4}{g_6} = \frac{J_{zz}}{3|J_{\pm}|}, \quad (3)$$

whereas the physical hexagon process appears only at third order. The artifact therefore diverges relative to

the physical process in exactly the perturbative limit where the cluster would otherwise be most trustworthy. Cubic-16 hosts 36 such winding four-cycles against 16 contractible hexagons; FCC-32 reduces but does not remove the problem, with 24 winding four-cycles, 32 contractible hexagons, and 96 wrapping hexagons. Due to the poor scaling of ED we cannot reach a cluster on which the four-site loop is absent.

In this Letter we remove the artifact by coupling microscopic spin transfers to an auxiliary source that records their net transport on the covering lattice—the infinite lattice obtained by unwrapping the periodic cell into translated copies—and retaining only the zero-transport component. Our first methodological result is that this projection needs neither a reference frame nor a spectrally isolated band: it is realized exactly by the Feshbach map onto the combinatorial ice manifold, and as a genuine Hermitian operator by a counterterm supported inside that manifold. Our central result concerns the one apparent blemish of that operator, the energy at which the counterterm is exact. We show that this energy is fixed by a self-consistency condition, that at its solution the construction is exact for the ground multiplet including the degeneracy which the transport-sector decomposition forces, and that solving the condition to high precision—rather than iterating it a few times—removes what we had previously mistaken for an intrinsic residue. We close by measuring what actually limits the whole family of constructions.

Transport character.—Periodic boundary conditions identify virtual processes that remain distinct on the covering lattice. In the elementary hexagonal process an S^+S^- operation creates a spinon–antispinon pair, which propagates around a contractible hexagon and annihilates. The spurious four-site process closes only through the periodic identification: on the covering lattice it is not a closed loop but an open string that transports one defect across a period before the pair recombines. Both connect ice states and have identical endpoints on the torus; their topology differs only on the cover.

A periodic cluster is specified by three period vectors, collected as the rows of a 3×3 matrix L , and site positions $\mathbf{r}_i$ in a chosen fundamental cell. For every oriented bond (i, j) we choose $\mathbf{n}_{ij} \in \mathbb{Z}^3$ such that the covering-lattice image of j connected to i is $\mathbf{R}_{j|i} = \mathbf{r}_j - \mathbf{n}_{ij}^T L$, so $\mathbf{n}_{ij} = 0$ for an intracell bond. For a configuration $\mathcal{C}$ lifted continuously to the cover we introduce the polarization coordinate

$$\mathbf{p}(\mathcal{C}) = \sum_{i=1}^N S_i^z(\mathcal{C}) \mathbf{R}_i, \quad (4)$$

the first spatial moment of the site-resolved Ising polarization. It introduces no physical degree of freedom and is not a dipole moment; its only role is to record how S^z is displaced by a sequence of microscopic operations. For

any sequence γ the accumulated transport is

$$\mathbf{q}_\gamma = 2 \sum_{\ell \in \gamma} \Delta \mathbf{p}_\ell. \quad (5)$$

The factor of two is a quantization requirement rather than a convention. With $S^z = \pm \frac{1}{2}$ the raw increments are half-integer multiples of the periods, and doubling promotes every closed ice-to-ice sequence to an integer vector in units of the cluster periods, as in $\mathbf{q}_\gamma = (0, 0, -1)$ of Fig. 1(i.b). Integrality is what makes $e^{-i\boldsymbol{\theta} \cdot \mathbf{q}_\gamma}$ a character of the source torus, and hence the average over $\boldsymbol{\theta}$ a genuine group-orthogonality sum, $(2\pi)^{-3} \int d^3\boldsymbol{\theta} e^{-i\boldsymbol{\theta} \cdot \mathbf{q}_\gamma} = \delta_{\mathbf{q}_\gamma, \mathbf{0}}$. A half-integer label would be *mutilated* by the same average rather than removed—a point we return to, because it is what ultimately bounds the method.

Since $\mathbf{q}$ is conserved by every physical ice-to-ice process, the winding-free effective Hamiltonian on the ice manifold is block diagonal in the transport sectors, and the projection is implemented by the mask

$$(\Delta[A])_{ab} = A_{ab} \delta_{\mathbf{q}_a, \mathbf{q}_b}, \quad (6)$$

which deletes exactly the boundary-winding matrix elements. On cubic-16 the 90 ice configurations fall into 25 such sectors, of dimensions

$$6 \times 1 + 12 \times 2 + 6 \times 8 + 1 \times 12 = 90. \quad (7)$$

Sectors related by a cubic operation carry identical spectra under any masked operator. The winding-free spectrum therefore has *exact* degeneracies, and Eq. (7) is what fixes them; this is used below.

Frame-free downfolding.—The construction requires an effective Hamiltonian on the ice manifold to which Eq. (6) can be applied. Previous implementations obtained one by projecting the exact low-energy band onto P_{ice} and orthonormalizing with the canonical polar isometry. That route requires the band to remain spectrally isolated and the projected Gram matrix $P_{\text{ice}} P P_{\text{ice}}$ to remain invertible, and we show below that both fail well before the couplings of experimental interest.

Neither is necessary. Splitting the Hilbert space with $P = P_{\text{ice}}$ and $Q = 1 - P$ and eliminating Q exactly gives the Feshbach map

$$F(z) = P H_0 P + P H_0 Q (z - Q H_0 Q)^{-1} Q H_0 P, \quad (8)$$

whose self-consistent roots $E = \text{eig } F(E)$ are exact eigenvalues of H_0 with nonzero ice weight. Nothing here refers to eigenvectors, to adiabatic continuation, or to a Gram matrix: P is combinatorial and exists at every coupling. Each sorted branch of F is monotone between poles of the self-energy, so every root below the continuum is located by guaranteed bisection; against the exact spectrum at $J_\pm/J_{zz} = -0.05$ the ninety unmasked roots reproduce the lowest ninety eigenvalues to $3 \times 10^{-13} J_{zz}$. The mask

of Eq. (6) is a pinching and preserves that monotonicity, so $\Delta[F(z)]$ inherits the same guarantee. We take the masked roots as the *reference* winding-free spectrum throughout.

The counterterm.—The masked map delivers the gauge sector but not the rest of the Hilbert space: it has no states above the band, so it yields neither the spinon anomaly nor any observable requiring a wavefunction. For those one needs an operator. Let $C = C^\dagger$ be supported inside the ice manifold, $C = PCP$. Then $Q(H_0 + C)Q = QH_0Q$ and $Q(H_0 + C)P = QH_0P$: the self-energy is untouched, and

$$F_{H_0+C}(z) = F_{H_0}(z) + C \quad \text{for every } z. \quad (9)$$

A counterterm living inside P has no matrix element reaching the spinon sector and therefore cannot dress itself; it enters the effective theory with exactly the coefficient chosen. Choosing

$$C = -(1 - \Delta)[F_{H_0}(z_0)] \quad (10)$$

makes $F_{H_0+C}(z_0) = \Delta[F_{H_0}(z_0)]$ exactly: the winding-free effective Hamiltonian is realized as the Feshbach map of a genuine Hermitian operator on the full Hilbert space. Every transporting process—four-cycles, wrapping hexagons and the longer transporting strings—is cancelled at once, with no loop-class enumeration and no perturbative hierarchy, at a cost of ninety linear solves per evaluation.

The anchor is not a parameter.—Equation (10) is exact at z_0 and approximate elsewhere, and this is the only respect in which the operator falls short of the map. The resolution is that z_0 is not free. It is determined by the Brillouin–Wigner condition

$$z_0 = E_0(H_0 + C(z_0)), \quad (11)$$

and at a solution of Eq. (11) the ground state of $H_0 + C$ sits exactly at the one energy where the counterterm is exact. The winding-free ground multiplet is therefore reproduced not approximately but exactly—and in particular with the degeneracy that Eq. (7) forces, which is a theorem about the mask rather than a numerical coincidence. We verify that the protected degeneracies of $\Delta[F(z_0)]$ hold to 6×10^{-7} of the band width at every coupling, and that the multiplicity is 6 throughout $J_\pm/J_{zz} \geq -0.22$, changing to 2 from -0.25 on as the sector structure reorganizes.

Everything then depends on how well Eq. (11) is solved, and this is the point we had previously missed. Iterating $z \mapsto E_0(H_0 + C(z))$ converges linearly at rate $\sim 1 - Z$, where Z is the ice weight introduced below; at $J_\pm/J_{zz} = -0.20$, where $Z \simeq 0.5$, that is a factor of two per expensive solve. A secant step on $g(z) = E_0(H_0 + C(z)) - z$ costs the same and converges superlinearly, reaching $|g| < 10^{-13}$ in five to eleven solves.

Tracking the splitting δ of the protected ground multiplet along that trajectory gives, at $J_\pm/J_{zz} = -0.20$,

$$\delta \simeq 1.5 |g| \quad (12)$$

over ten decades, from $\delta = 4 \times 10^{-2}$ at $|g| = 6 \times 10^{-2}$ down to $\delta = 1 \times 10^{-12}$ at $|g| = 5 \times 10^{-15}$ [13].

The consequence is a correction to our own earlier reading of this construction. An under-converged anchor splits the protected multiplet; by the Schottky relation a splitting δ manufactures a heat-capacity anomaly at $T \approx \delta/2.4$, and it simultaneously breaks the entropy sum rule $\int_0^\infty (C/NT) dT = N^{-1} \ln(2^N/g_0)$, because levels split too finely to release entropy are also not exactly degenerate enough to count as frozen. Both effects are real and we had reported them—a spurious low-temperature peak at every coupling, and a 10% sum-rule violation at weak coupling—as an intrinsic “anchoring residue” requiring a repair. Equation (12) shows they are neither intrinsic nor in need of repair: with Eq. (11) solved to 10^{-13} the worst splitting anywhere in the sweep is $\sim 10^{-12} J_{zz}$, the anomaly it could produce lies twelve orders of magnitude below g_6 , the sum rule is satisfied to 5×10^{-7} at every coupling with the measured g_0 , and no maximum of $C(T)$ appears at the residual scale.

A third maximum does appear, but for a different and physical reason, and only for $|J_\pm/J_{zz}| \geq 0.22$. The protected multiplet is separated from the rest of the band by a genuine gap, which produces its own Schottky anomaly; at weak coupling that anomaly coincides with the ring anomaly and at strong coupling it resolves from it. It is present in the exact masked reference, whose gap the counterterm reproduces to 3%, so it is a property of cubic-16’s twenty-five-block structure rather than of the method. This is precisely what the unconverged variant obscured: it produced a spurious branch at *every* coupling, including those where the physical one is merged.

We also tested the repair we had proposed, and it makes things worse. The degeneracy pattern of $\Delta[F(z_0)]$ is exact, so one can read the protected multiplets off it and average the counterterm spectrum within each; averaging is the unique trace-preserving choice. Applied to the converged spectrum this degrades the ring-exchange scale from 0.66% to 1.16% of the reference at $J_\pm/J_{zz} = -0.20$, and from 1.86% to 2.65% at -0.22 . The reason identifies where the counterterm’s real error lives: Eq. (10) is exact at z_0 , so $\Delta[F(z_0)]$ describes the bottom of the band well and the top badly—by mid-band at -0.20 its levels overshoot the reference by more than a factor of two—and while its degeneracy pattern is exact its eigenvectors are not, with an overlap purity of only 0.57. The counterterm’s residual error is the ordinary $O(1-Z)$ energy dependence of the exact downfolded theory spread through the upper band, and the correct response is to place the exact point where the physics is decided and then quote Z , not to post-process the spectrum.

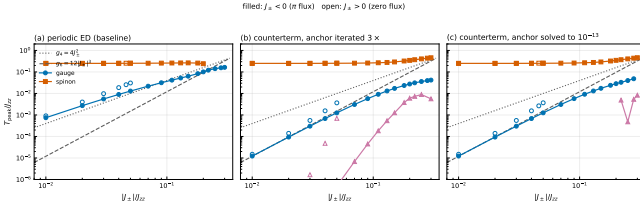


FIG. 2. Every maximum of $C(T)$ tracked across coupling against the two candidate scales: the baseline, and the counterterm before and after the anchor is converged. Circles: the ring (for periodic ED, gauge) anomaly. Triangles: transport-sector anomalies. Squares: the spinon anomaly. Filled symbols are the π -flux side, open the zero-flux side. Periodic ED tracks g_4 throughout; the winding-free construction tracks g_6 . Panel (b) is the counterterm with the anchor iterated three times, as in an earlier version of this work: a sector branch runs along the bottom at every coupling. Panel (c) is the same construction with Eq. (11) solved to 10^{-13} : that branch is gone, and the sector anomaly that remains for $|J_{\pm}|/J_{zz} \geq 0.22$ is the physical multiplet gap, present in the exact reference too.

Results.—We computed the complete 65536-level spectrum—every magnetization sector, translation resolved—at 22 couplings spanning $-0.30 \leq J_{\pm}/J_{zz} \leq 0.05$, and evaluated $C(T)$ from each. The counterterm modifies only the $S_{\text{tot}}^z = 0$ block, so the remaining sectors are shared with periodic ED exactly. Every peak is labelled by the number of thermally active levels that produce it; the checks (anchor residual, protected degeneracies, entropy sum rule, temperature-grid refinement, mask convergence) are reported in the Supplemental Material [13].

The separation of scales is unambiguous (Fig. 2). The winding-free anomaly settles onto the ring-exchange scale as $J_{\pm} \rightarrow 0$, as it must since perturbation theory becomes exact there: $1.024 g_6$ at $J_{\pm}/J_{zz} = -0.01$ and $1.245 g_6$ at $+0.01$. Neither is exactly g_6 , nor should it be—a Schottky maximum sits at $\Delta/2.4$ only for a two-level system, and the prefactor depends on how the ninety band levels are distributed, which differs between the frustrated and unfrustrated signs. Ordinary periodic ED instead approaches $1.87 g_4$ and departs from g_6 as $g_4/g_6 \propto 1/|J_{\pm}|$, by a factor 62 at $J_{\pm}/J_{zz} = -0.01$. Fitting $T_{\text{peak}} \propto |J_{\pm}|^n$ over $|J_{\pm}| \leq 0.03$, with peak positions refined below the temperature-grid spacing, gives $n = 1.99$ for periodic ED against the ideal 2 of g_4 , and $n = 3.10$ for the winding-free construction against the ideal 3 of g_6 .

Two comparisons with the frame-free reference are worth separating. The sharpest is the ground energy, where the fixed point makes a parameter-free prediction: it agrees with the lowest exact masked root to between 10^{-15} and $7 \times 10^{-13} J_{zz}$ at every coupling. The ring anomaly agrees to 0.18% or better throughout $|J_{\pm}|/J_{zz} \leq 0.18$, degrading to 0.66% at -0.20 and 1.86% at -0.22 . Individual band *levels* are far less accu-

rate than either: the largest deviation grows from 0.12% of the band width at -0.01 to 16.3% at -0.20 . That hierarchy—an exact ground energy, a per-cent-accurate anomaly, and levels an order of magnitude worse—is the signature of a construction exact at one energy and degrading away from it, and the anomaly survives it because its position is set by how the band is distributed rather than by where any individual level sits. Beyond -0.22 the reference loses branches; they are not missing states but real-energy bound states in the continuum, protected by transport superselection, that the root finder cannot reach [13]. The spinon anomaly is essentially common to both constructions at weak coupling ($0.25307 J_{zz}$ against 0.25282 for periodic ED at -0.05), as it should be since neither modifies the spinon sector; it separates at strong coupling (0.347 against 0.241 at -0.20) not because that sector changed but because removing the g_4 network compresses the gauge band underneath it, and the position of a maximum depends on the whole spectrum.

One prediction of the construction is worth isolating, because it is qualitative and requires no reference. The transport label of Eq. (5) is a linear function of the four sublattice magnetizations, so on cubic-16 the $\mathbf{k} = 0$ operator $P_{\text{ice}} S_0^z P_{\text{ice}}$ is *exactly* constant within each transport sector—one value per sector, to machine precision. It therefore commutes with the mask and is a function of the conserved label, and cannot connect the winding-free ground multiplet to anything: the winding-free longitudinal response at $\mathbf{k} = 0$ vanishes identically, measured as 5×10^{-30} against 0.333 for the same quantity with the mask removed. The entire $\mathbf{k} = 0$ longitudinal weight that ordinary periodic exact diagonalization reports on this cluster is boundary artifact.

At the sign-free benchmark $(J_{\pm}, J_{\pm\pm})/J_{zz} = (0.046, 0)$ the winding-free maximum lies at $2.27 g_6$ against $0.94 g_6$ from QMC in the thermodynamic limit [10], while ordinary ED gives $22 g_6$: the construction agrees with QMC on the *scale* but differs by a factor of two, as does the polar pullback in the other direction ($1.24 g_6$). That spread is not a test of the method—the same peak moves by a factor of three between cubic-16 and FCC-32, so a 16-site cluster cannot resolve a factor of two against the thermodynamic limit [13]. The sharp validations are internal: the exact ground energy, the 0.2% ring agreement, and the g_6 exponent.

What limits the method.—All constructions degrade, and they do so for one shared reason. The Supplemental Material [13] collects every failure metric beside the residue $Z = [1 - \langle \partial_z \Sigma \rangle]^{-1}$, the fraction of the true eigenstate carried by the ice manifold. The Gram minimum of the polar route *is* Z to within a few per cent; the disagreement between constructions is $0.4(1 - Z)/Z$; root loss in the reference begins at $Z \approx 0.37$. Across the range studied Z falls from 0.99 to 0.26, because virtual spinon pairs proliferate as $\langle n_{\text{pairs}} \rangle \approx 2N(J_{\pm}/J_{zz})^2$.

The conflict is structural. The mask requires an integer transport label, and the label is an integer only on the ice manifold, since a single spinon hop displaces $\mathbf{p}$ by a quarter period and would be mutilated rather than removed by the character average. The model space is therefore forced to be the ice manifold, while describing the physics at large $|J_{\pm}|$ requires precisely the spinon admixture the mask cannot accommodate. One cannot enlarge the model space to improve Z without destroying the label the construction rests on.

What does *not* happen is worth stating, since we had previously suspected otherwise. Nothing singular occurs to the cluster: the ground state stays nondegenerate, the gap grows monotonically, and every dressing measure is smooth—in particular there is no feature at $J_{\pm}/J_{zz} = -0.23$, where an earlier reading of a coherence diagnostic had placed a boundary. The π -flux side does not go transverse-ordered, and the winding-free gauge sector does not stop existing: read off the exact map it remains a ring-exchange model with negligible potential ($|V/g| \leq 0.012$, excluding a Rokhsar–Kivelson freezing scenario) whose amplitude grows in absolute terms while becoming progressively longer ranged [13]. The spinon sector retains its own finite-size error, which we measure by twisting the boundary condition: the spinon threshold moves by 3.8% of its gap at -0.05 and 24.8% at -0.30 , against an ice-sector response of several times g_4 —an ordinary finite-size effect, not a second copy of the winding artifact, because a spinon path around the box has a bulk counterpart at the same order whereas the winding four-cycle has none.

Outlook.—The self-consistent counterterm is, on the evidence assembled here, the construction to use: it needs no reference frame, no isolated band and no fitted correction, satisfies the entropy sum rule identically, reproduces the exact masked ground energy to machine precision and the ring anomaly to 0.2%, produces both thermodynamic anomalies and all observables from a single diagonalization, and continues past the coupling at which the reference loses branches. Its remaining limitations are quantified rather than hidden: an $O(1 - Z)$ error in the upper band, an untouched spinon sector, and a cluster whose winding-free blocks are at most twelve-dimensional—narrow enough that by $J_{\pm}/J_{zz} = -0.30$ the band has grown as wide as the spinon scale and the ring anomaly ceases to be a separate maximum at all. The last of these is the one that matters most. Comparing cubic-16 with FCC-32 at matched truncation moves the ring anomaly by a factor of 2.4 to 3.5—larger than every other uncertainty in the calculation combined. On FCC-32 the transport sectors reach dimension 396 and the construction applies unchanged; that is where the quantitative photon thermodynamics of the π -flux can-

didates should be computed.

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